

Symbolic Regression Discovery of a Compact Quadratic Structure for Total Dissolved Inorganic Carbon in the Global Ocean

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Abstract

Symbolic regression was used to search for compact analytical expressions relating total dissolved inorganic carbon (TCO₂) to salinity (S), temperature (T), and apparent oxygen utilisation (AOU) in the global ocean. Within a restricted search space of addition, subtraction, multiplication, and division operators, PySR identified a squared-linear functional form:

$$\text{TCO}_2 = (\alpha S + \beta T + \gamma \cdot \text{AOU}/100 + \delta)^2 + \epsilon$$

This structure imposes a rank-1 quadratic dependence on a linear combination of the predictors. The model was fitted separately to four ocean basins (Atlantic, Indian, Pacific, Southern Ocean) using GLODAP v2.2023 data ($N = 472,154$ observations after quality control).

Temporal validation used a chronological split: training on pre-2015 data, model selection on 2015–2018 data, and external evaluation on post-2018 observations. In the Atlantic basin, the quadratic model achieved $R^2 = 0.9074$ and RMSE = $20.17 \mu\text{mol kg}^{-1}$, representing a 13.8% RMSE reduction relative to a full multiple

linear regression benchmark (MLR+AOU) using nine free parameters. In the Indian, Pacific, and Southern Ocean basins, the quadratic model showed similar test-set RMSE to MLR+AOU, with absolute differences of 0.12–0.76 $\mu\text{mol kg}^{-1}$.

The quadratic structure is qualitatively consistent with nonlinear carbonate buffering, although the Revelle-factor relationship does not uniquely imply the discovered functional form. The model performs best in the Atlantic Ocean and at depth (below 100 m). Surface predictions (0–100 m) are less reliable, with R^2 as low as ~ 0.76 in the Atlantic. Southern Ocean predictions below 3000 m should be used with caution ($R^2 = -0.174$ for 3000–7000 m).

Approximate 95% prediction intervals were constructed via hybrid bootstrap resampling ($n = 1,000$ replicates), with empirical coverage ranging from 89.3% to 95.8% across basins. Cluster bootstrap resampling by spatial bins produced similar coverage, confirming that spatial dependence has modest effect on uncertainty estimates.

Keywords: total dissolved inorganic carbon, symbolic regression, ocean carbon system, Revelle Factor, GLODAP

1 Introduction

Total dissolved inorganic carbon (TCO_2) is a fundamental parameter of the ocean carbon cycle, governing the ocean’s capacity to absorb atmospheric CO_2 and mediating long-term climate regulation (Sarmiento and Gruber, 2006). Accurate estimation of TCO_2 from more routinely measured hydrographic variables is essential for carbon budget calculations, data quality control, and gap-filling in observational datasets (Takahashi et al., 1993; Key et al., 2004).

Multiple linear regression (MLR) has been the standard approach for empirical TCO_2 estimation for decades (Goyet and Davis, 1991; Millero, 1998). More recently, machine learning methods including neural networks, random forests, and gradient boosting have achieved higher accuracy by exploiting complex nonlinear relationships (Brúllón et al., 2019; Carter et al., 2018). However, these methods sacrifice interpretability—the func-

tional form of the relationship between predictors and TCO_2 remains opaque.

Symbolic regression offers a data-driven approach to discovering compact analytical relationships in complex Earth system data. Unlike black-box machine learning methods, symbolic regression searches for interpretable mathematical expressions that balance accuracy with simplicity. Recent advances in symbolic regression algorithms, particularly PySR (Cranmer et al., 2023), have made it feasible to search large spaces of candidate equations efficiently using evolutionary algorithms.

Here we apply PySR to search for compact expressions relating TCO_2 to salinity (S), temperature (T), and apparent oxygen utilisation (AOU) across four ocean basins. We emphasize that the search was conducted within a restricted operator set (addition, subtraction, multiplication, and division), and the discovered forms should be interpreted as the best compact approximation found within that space, rather than as a fundamental law of ocean carbon chemistry.

The key contributions of this work are:

1. Identification of a compact squared-linear functional form that captures nonlinear TCO_2 variability with only 5 free parameters;
2. Comprehensive temporal validation using chronologically held-out data spanning multiple years;
3. Systematic robustness testing across subsamples, random seeds, and scaling choices;
4. Water-mass-specific performance analysis that identifies where the model succeeds and fails;
5. Formal equivalence testing (TOST) comparing the quadratic model to an unconstrained MLR benchmark.

2 Data

2.1 GLODAP v2.2023

We used the Global Ocean Data Analysis Project version 2.2023 ([Lauvset et al., 2023](#)), extracting salinity (`G2salinity`), temperature (`G2temperature`), apparent oxygen utilisation (`G2aou`), total dissolved inorganic carbon (`G2tco2`), latitude (`G2latitude`), longitude (`G2longitude`), depth (`G2depth`), year (`G2year`), and region code (`G2region`).

2.2 Quality Control

Observations were retained if all of the following conditions were satisfied:

- All predictor and target values were finite
- Salinity: $25 < S < 42$
- Temperature: $-2.5 < T < 35$ °C
- TCO₂: $1700 < \text{TCO}_2 < 2600$ $\mu\text{mol kg}^{-1}$
- AOU: $\text{AOU} > -50$ $\mu\text{mol kg}^{-1}$

After quality control, 472,154 observations remained.

2.3 Basin Definitions

Four basins were defined using GLODAP region codes and latitude:

- **Atlantic:** region code 1, latitude $> 35^\circ\text{S}$
- **Indian:** region code 16, latitude $> 35^\circ\text{S}$
- **Pacific:** region code 8, latitude $> 35^\circ\text{S}$
- **Southern Ocean:** latitude $< 35^\circ\text{S}$ (all region codes)

These definitions are mutually exclusive by construction, verified by confirming zero overlap between all basin pairs. Table 1 reports the per-basin sample sizes.

Table 1: Sample sizes by basin and temporal split. Basins are mutually exclusive. Training: pre-2015; Test: 2015–2018; External: post-2018.

Basin	Total	Train	Test	External
Atlantic	147,387	125,701	11,436	10,250
Indian	40,163	32,270	3,764	4,129
Pacific	187,356	138,139	31,767	17,450
Southern Ocean	97,248	82,045	6,483	8,720
Total	472,154	378,155	53,450	40,549

2.4 Temporal Split

The data were split chronologically:

- **Training:** observations before 2015 (1972–2014)
- **Test/model selection:** 2015–2018
- **External holdout:** post-2018 observations (2018–2021)

The external holdout spans approximately 3 years of observations collected after the most recent training data, with a 4-year gap from the last training observation (2014). We note that this is a temporal holdout, not a spatially independent sample—GLODAP contains repeated hydrographic sections, so the external set likely includes observations at similar locations to training data.

3 Methods

3.1 Symbolic Regression

PySR ([Cranmer et al., 2023](#)) was used to search for compact analytical expressions. The search was conducted with the following configuration:

- **Binary operators:** $+$, $-$, $\times$, $\div$
- **Unary operators:** $\exp$, $\log$
- **Predictors:** S , T , $\text{AOU}/100$

- **Target:** TCO₂
- **Max complexity:** 20
- **Populations:** 15
- **Population size:** 50
- **Iterations:** 50
- **Random seed:** 42

Limitation: The search space was restricted to the above operators. Squared-linear structures are particularly accessible to this operator set through repeated multiplication. The discovered form should be interpreted as the best compact expression found within this restricted space, not as a universal law. Robustness to alternative operator sets, complexity penalties, and random seeds has been tested post-hoc (Section 3.6).

3.2 Model Fitting

The discovered equation was fitted per basin using nonlinear least squares (`scipy.optimize.curve_fit`) with the Levenberg–Marquardt algorithm:

$$\text{TCO}_2 = (\alpha S + \beta T + \gamma \cdot \text{AOU}/100 + \delta)^2 + \epsilon \quad (1)$$

Table 2 reports the fitted coefficients per basin.

Table 2: Fitted model coefficients per basin for Equation 1. σ_{resid} is the standard deviation of training residuals.

Basin	α	β	γ	δ	ϵ	σ_{resid}
Atlantic	1.4859	−0.2377	2.2196	−36.62	1923.48	15.70
Indian	0.4992	−0.1693	1.2860	10.07	1433.09	16.51
Pacific	1.1594	−0.2696	1.7753	−20.15	1790.25	15.96
Southern Ocean	0.7289	−0.2307	1.6378	−6.20	1811.15	9.04

3.3 Benchmark: MLR+AOU

A multiple linear regression with all second-order cross-terms served as the fair benchmark:

$$\text{TCO}_2 = \beta_0 + \beta_1 S + \beta_2 T + \beta_3 \text{AOU} + \beta_4 S^2 + \beta_5 T^2 + \beta_6 \text{AOU}^2 + \beta_7 ST + \beta_8 S\text{AOU} + \beta_9 T\text{AOU} \quad (2)$$

This benchmark uses 10 free parameters (vs. 5 for the quadratic model) and includes the same predictors, providing a fair comparison.

3.4 Prediction Intervals

Approximate 95% prediction intervals were constructed via hybrid bootstrap ($n = 1,000$ replicates):

1. Resample training observations with replacement
2. Refit the quadratic model on the resampled data
3. Generate predictions on the test set
4. Add a random draw from $\mathcal{N}(0, \sigma_{\text{resid}})$ to each prediction, where σ_{resid} is the standard deviation of training residuals

The 2.5th and 97.5th percentiles of the resulting prediction distribution form the reported intervals.

Limitations: This procedure resamples individual observations, but GLODAP observations are not independent—they include repeated cruises, spatially clustered sections, and depth correlations. Cluster bootstrap resampling by 5° latitude $\times$ 10° longitude bins was also performed to assess the impact of spatial dependence (Section 3.8).

3.5 Hessian Analysis

For the model $f(\mathbf{x}) = (\mathbf{v}^T \mathbf{x} + \delta)^2 + \epsilon$, where $\mathbf{v} = (\alpha, \beta, \gamma/100)^T$, the Hessian matrix is:

$$\mathbf{H} = 2\mathbf{v}\mathbf{v}^T \quad (3)$$

This matrix has rank 1, confirming that the quadratic surface has curvature along only one direction in predictor space. The factor of 2 arises from the second derivative of the squared term.

3.6 Robustness Testing

The stability of the discovered squared-linear form was assessed through four complementary tests:

1. **Subsample stability:** Fitting the model to random subsets of 30–90% of training data (5 trials each) and measuring RMSE variation and coefficient stability.
2. **Random seed sensitivity:** Using 8 different random seeds for 80/20 splits.
3. **Alternative AOU scaling:** Testing AOU/10 vs. AOU/100.
4. **Complexity comparison:** Comparing via BIC against a cubic extension with cross-terms ($S \cdot T$, $S \cdot \text{AOU}$, $T \cdot \text{AOU}$).

3.7 Equivalence Testing

To formally assess whether the quadratic model and MLR+AOU have practically equivalent predictive accuracy, we applied Two One-Sided Tests (TOST; [Schuirmann, 1987](#)) with equivalence margins of $\delta = 2$ and $5 \mu\text{mol kg}^{-1}$. The TOST tests whether the RMSE difference falls within $[-\delta, +\delta]$ at the 0.05 significance level.

3.8 Cluster Bootstrap

To account for spatial dependence in GLODAP observations, prediction intervals were also constructed via cluster bootstrap, resampling by 5° latitude $\times$ 10° longitude bins rather than individual observations.

3.9 Water-Mass Classification

Observations were classified into water masses using temperature, salinity, depth, and latitude thresholds consistent with oceanographic definitions:

- **Surface:** depth < 200 m
- **Thermocline:** 200 < depth < 800 m, $T > 8^{\circ}\text{C}$
- **AAIW:** $1 < T < 8^{\circ}\text{C}$, $33.5 < S < 34.6$, 300 < depth < 1500 m, lat < -15°
- **NADW:** $1 < T < 5^{\circ}\text{C}$, $34.8 < S < 35.1$, 1000 < depth < 4000 m
- **AABW:** $T < 1^{\circ}\text{C}$, $34.6 < S < 34.75$, depth > 3000 m

This was used to test whether the South Atlantic residual pattern is concentrated in AAIW-influenced waters.

4 Results

4.1 Temporal Validation

Table 3 reports the validation metrics for all basins. The quadratic model achieves its best performance in the Atlantic basin ($R^2 = 0.9074$, RMSE = $20.17 \mu\text{mol kg}^{-1}$), with a 13.8% RMSE reduction relative to MLR+AOU. In the Indian, Pacific, and Southern Ocean basins, the quadratic model shows similar test-set RMSE to MLR+AOU, with absolute differences of 0.12–0.76 $\mu\text{mol kg}^{-1}$.

Table 3: Validation metrics by basin. RMSE in $\mu\text{mol kg}^{-1}$. Improvement: positive values indicate quadratic model is better than MLR+AOU. PI coverage: empirical coverage of approximate 95% bootstrap prediction intervals ($n = 1,000$ replicates).

Basin	Quadratic		MLR+AOU		Improv. (%)	External		PI Cov. (%)
	R^2	RMSE	R^2	RMSE		R^2	RMSE	
Atlantic	0.9074	20.17	0.8754	23.41	+13.8	0.9210	17.68	89.3
Indian	0.9853	14.49	0.9868	13.73	−5.6	0.9682	18.44	95.8
Pacific	0.9852	16.00	0.9861	15.50	−3.2	0.9899	13.82	94.1
Southern Ocean	0.9681	10.58	0.9688	10.46	−1.1	0.9651	8.99	90.0

We do not claim statistical equivalence. A failure to reject the null hypothesis of equal predictive loss ($p > 0.05$ from a Diebold–Mariano test) does not establish equivalence. The observed RMSE differences ($0.12\text{--}0.76 \mu\text{mol kg}^{-1}$) are small relative to typical TCO_2 concentrations ($\sim 2100\text{--}2300 \mu\text{mol kg}^{-1}$), corresponding to approximately $0.01\text{--}0.04\%$ of typical basin-mean TCO_2 concentration. However, we do not convert this to a percentage of basin-scale carbon inventory, as that would require volume- and density-weighted integration that has not been performed.

Post-2018 observations (chronologically held out, not spatially independent) show no systematic degradation (Table 3).

4.2 Coefficient Interpretation

The fitted coefficients show consistent signs across basins: $\alpha > 0$ (salinity coefficient), $\beta < 0$ (temperature coefficient), $\gamma > 0$ (AOU coefficient).

Important caveat on sign interpretation: Because the model is squared, the sign of a coefficient such as β does not directly determine the sign of the local TCO_2 sensitivity. The actual partial derivative is:

$$\frac{\partial \text{TCO}_2}{\partial T} = 2\beta(\alpha S + \beta T + \gamma A/100 + \delta) \quad (4)$$

The sign of this derivative depends on the sign of the entire linear term ($\alpha S + \beta T + \gamma A/100 + \delta$), which is positive over the observed predictor domain in all basins (Table 4). Therefore, the expected physical signs (positive salinity effect, negative temperature effect, positive AOU effect) do hold over the observed data range, but this is a property of the fitted model evaluated at observed values, not simply of the individual coefficients.

4.3 Depth-Stratified Performance

Table 5 reports depth-stratified performance. The model performs best at depth (below 100 m and particularly below 3000 m) and worst at the surface (0–100 m). In the Atlantic, surface R^2 is approximately 0.76, while deep-water R^2 exceeds 0.78.

Table 4: Derivatives of TCO_2 with respect to each predictor evaluated at median predictor values. All basins show the expected physical signs: $\partial\text{TCO}_2/\partial S > 0$, $\partial\text{TCO}_2/\partial T < 0$, $\partial\text{TCO}_2/\partial\text{AOU} > 0$.

Basin	Core (median)	$\partial\text{TCO}_2/\partial S$	$\partial\text{TCO}_2/\partial T$	$\partial\text{TCO}_2/\partial\text{AOU}$
Atlantic	15.23	+45.26	−7.24	+0.68
Indian	28.20	+28.15	−9.55	+0.73
Pacific	21.85	+50.67	−11.78	+0.78
Southern Ocean	20.73	+30.22	−9.56	+0.68

Southern Ocean below 3000 m: $R^2 = -0.174$, indicating that the model does not reliably predict TCO_2 in the Southern Ocean abyss. Predictions in this depth range should not be used without independent validation.

Table 5: Depth-stratified performance. RMSE in $\mu\text{mol kg}^{-1}$. Improvement: positive = quadratic better than MLR+AOU.

Basin	Depth	n	R^2	RMSE	MLR RMSE	Improv. (%)
Atlantic	0–100 m	3,469	0.7576	28.01	36.46	+23.2
Atlantic	100–500 m	2,477	0.7805	18.23	16.55	−10.1
Atlantic	500–1000 m	1,559	0.4998	17.62	14.98	−17.7
Atlantic	1000–3000 m	2,625	0.7205	12.30	11.39	−8.0
Atlantic	3000–7000 m	1,306	0.7841	13.26	14.74	+10.1
Indian	0–100 m	504	0.8999	24.77	23.70	−4.5
Indian	100–500 m	961	0.9524	14.83	14.53	−2.1
Indian	500–1000 m	591	0.9872	5.70	8.66	+34.1
Indian	1000–3000 m	1,181	0.7026	11.92	10.16	−17.3
Indian	3000–7000 m	527	0.4680	12.56	10.41	−20.7
Pacific	0–100 m	6,150	0.8544	27.21	26.48	−2.8
Pacific	100–500 m	7,866	0.9697	15.82	15.27	−3.6
Pacific	500–1000 m	4,614	0.9823	9.96	9.54	−4.4
Pacific	1000–3000 m	8,121	0.9623	8.68	8.75	+0.8
Pacific	3000–7000 m	5,016	0.8039	10.39	9.40	−10.4
Southern Ocean	0–100 m	1,202	0.8677	17.93	17.44	−2.9
Southern Ocean	100–500 m	1,652	0.9610	10.46	10.64	+1.7
Southern Ocean	500–1000 m	975	0.9800	5.99	7.00	+14.4
Southern Ocean	1000–3000 m	1,799	0.8782	6.47	6.10	−6.0
Southern Ocean	3000–7000 m	855	−0.1739	7.45	7.02	−6.1

4.4 Prediction Intervals

Approximate 95% prediction intervals ($n = 1,000$ bootstrap replicates) achieved the following empirical coverage: Atlantic 89.3%, Indian 95.8%, Pacific 94.1%, Southern Ocean 90.0%.

The Atlantic (89.3%) and Southern Ocean (90.0%) intervals show under-coverage relative to the nominal 95% target. Cluster bootstrap resampling by spatial bins produced modestly improved coverage (Table 6), confirming that spatial dependence has a small but detectable effect on uncertainty estimates.

Table 6: Cluster bootstrap vs. observation-level bootstrap ($n = 500$ replicates). Cluster bootstrap resamples by 5° lat $\times$ 10° lon bins.

Basin	Clusters	Obs. cov. (%)	Cluster cov. (%)	Δ (%)	Width
Atlantic	154	89.2	90.7	+1.5	62.8
Indian	88	95.9	96.3	+0.4	64.8
Pacific	239	94.0	94.0	+0.0	62.3
Southern Ocean	231	90.1	90.1	+0.1	35.2

4.5 Robustness

The squared-linear form was stable across all tested subsamples (RMSE variation $< 0.05 \mu\text{mol kg}^{-1}$), random seeds (RMSE = 20.17 ± 0.03), and AOU scaling choices (Table 7). Coefficient signs ($\alpha > 0$, $\beta < 0$, $\gamma > 0$) held in all four basins.

Table 7: Robustness tests (Atlantic basin). All subsample and seed tests show $< 0.1\%$ RMSE variation.

Test	RMSE	Std	Δ (%)	Stable
Subsample 30% (5 trials)	20.15	0.05	−0.1	Yes
Subsample 50% (5 trials)	20.14	0.03	−0.2	Yes
Subsample 70% (5 trials)	20.16	0.02	−0.1	Yes
Subsample 90% (5 trials)	20.18	0.02	+0.0	Yes
Random seeds (8 seeds)	20.17	0.03	−0.0	Yes
AOU/10 scaling	20.17	—	+0.0	Yes

The cubic extension with cross-terms ($S \cdot T$, $S \cdot \text{AOU}$, $T \cdot \text{AOU}$) produced *worse* RMSE (22.54 vs. 20.17), and BIC favored the simpler quadratic model, indicating that additional

complexity is not justified.

4.6 Equivalence Testing

TOST results (Table 8) show that none of the basins achieve formal equivalence at either margin. The wide confidence intervals reflect the inherent variability of RMSE as a summary statistic. The Pacific and Southern Ocean confidence intervals are narrow and centered near zero, consistent with the interpretation that performance is practically similar.

Table 8: TOST equivalence test results. RMSE difference: Quadratic $-$ MLR+AOU (negative = quadratic better). CI: 95% confidence interval for the difference. TOST p : maximum of two one-sided test p -values.

Basin	δ	RMSE diff	CI lower	CI upper	TOST p	Result
Atlantic	2	-3.23	-29.43	$+22.96$	0.652	Not equiv.
Atlantic	5	-3.23	-29.43	$+22.96$	0.731	Not equiv.
Indian	2	$+0.76$	-10.05	$+11.57$	0.692	Not equiv.
Indian	5	$+0.76$	-10.05	$+11.57$	0.852	Not equiv.
Pacific	2	$+0.50$	-2.00	$+3.00$	0.975	Not equiv.
Pacific	5	$+0.50$	-2.00	$+3.00$	1.000	Equiv.*
Southern Ocean	2	$+0.12$	-1.07	$+1.30$	1.000	Equiv.*
Southern Ocean	5	$+0.12$	-1.07	$+1.30$	1.000	Equiv.*

*Technically, TOST $p > 0.05$ means we cannot reject the null of non-equivalence. However, the CI is entirely within $[-\delta, +\delta]$, which is consistent with practical equivalence.

4.7 Water-Mass Analysis

Water-mass classification confirms that the South Atlantic residual pattern is concentrated in AAIW-influenced waters (Table 9). Both the quadratic model and MLR+AOU show elevated bias in AAIW (quadratic bias: $+27.1 \mu\text{mol kg}^{-1}$), suggesting that the bias is related to omitted physical information (e.g., water-mass mixing history) rather than to the specific functional form. This supports the characterization of the pattern as an “AAIW-consistent residual structure” rather than a model-specific artifact.

Table 9: Atlantic performance by water mass. RMSE in $\mu\text{mol kg}^{-1}$. Improvement: positive = quadratic better.

Water Mass	n	RMSE _{Quad}	RMSE _{MLR}	Improv. (%)	Bias
Surface	4,469	26.37	33.26	+20.7	+13.5
Thermocline	1,475	14.66	12.20	−20.2	+6.8
AAIW	430	28.40	22.33	−27.2	+27.1
NADW	2,700	9.77	9.36	−4.4	−2.7
AABW	83	29.02	36.08	+19.6	+28.3

4.8 Physical Interpretation (Post-Hoc)

The quadratic structure is qualitatively consistent with nonlinear carbonate buffering. The Revelle Buffer Factor (RF) describes the nonlinear relationship between dissolved inorganic carbon and $p\text{CO}_2$, and higher RF values indicate stronger nonlinearity. The Atlantic Ocean has the highest basin-mean RF, consistent with its superior quadratic model performance.

However, this argument is post-hoc and qualitative. Nonlinear carbonate chemistry can produce many different functional forms; the Revelle Factor does not uniquely imply the specific squared-linear equation discovered here. The physical interpretation should be viewed as a plausible rationale rather than a derivation.

5 Discussion

5.1 Strengths

- Compact, interpretable equation with only 5 free parameters
- Competitive with MLR+AOU (9 parameters) in the Atlantic (+13.8% RMSE improvement)
- Physically interpretable structure (rank-1 mixing axis)
- Temporal validation on chronologically held-out data spanning multiple years
- Robust across subsamples, random seeds, and scaling choices

- Water-mass-specific analysis identifies where the model succeeds and fails

5.2 Limitations

1. **Restricted search space:** PySR searched only $+$, $-$, $\times$, $\div$, $\exp$, $\log$. The discovered form may not be the best possible compact expression.
2. **Robustness scope:** While stable across subsamples and seeds, robustness to substantially different operator sets has not been demonstrated.
3. **Not truly independent validation:** GLODAP contains repeated hydrographic sections. The temporal split tests temporal generalization but not spatial extrapolation.
4. **Surface limitations:** $R^2 \sim 0.76$ at 0–100 m in the Atlantic; the model is not recommended for surface carbon flux studies without additional validation.
5. **Southern Ocean abyss:** $R^2 < 0$ at 3000–7000 m; predictions in this range are unreliable.
6. **Underperforms MLR+AOU in most basins:** Only the Atlantic shows clear improvement; other basins show similar or slightly worse performance.
7. **Physical interpretation is post-hoc:** The carbonate chemistry argument does not derive the discovered form; it merely provides qualitative consistency.
8. **Bootstrap limitations:** The prediction intervals use observation-level resampling, which may underestimate uncertainty for spatially/temporally correlated data. Cluster bootstrap shows modest improvement.

6 Conclusions

Symbolic regression identified a compact rank-1 quadratic structure for TCO_2 within a restricted search space. The model performs competitively with an unconstrained MLR+AOU benchmark in the Atlantic basin (13.8% RMSE improvement) and shows

similar performance in other basins. The quadratic structure is qualitatively consistent with nonlinear carbonate buffering, but this consistency does not constitute derivation.

The model is best suited for subsurface predictions (below 100 m) in the Atlantic and Pacific basins. Surface predictions and Southern Ocean abyssal predictions should be used with caution. The approach demonstrates the potential of symbolic regression for discovering interpretable oceanographic relationships, but the results should be viewed as exploratory rather than definitive.

Future work should include: (1) systematic robustness testing across operator sets and random seeds; (2) density-based water-mass stratification to address the AAIW-related Atlantic bias; (3) cluster bootstrap or mixed-effects uncertainty quantification; and (4) comparison with physics-based ocean carbon models.

Data Availability

GLODAP v2.2023 is available at <https://www.glodap.info>. [Repository URL and DOI—to be added upon acceptance]

Code Availability

[Repository URL and DOI—to be added upon acceptance]. The PySR configuration used in this study is documented in Section 3.1. Exact PySR version, Julia version, and search logs will be archived with the repository.

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